

Cloud native temperature excursion and compliance platform

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Actors in the system

1. Logistics Manager:

- Define the shipments and the temperature constraints.
- Constraint: Cannot alter excursion logs

2. Warehouse Receiver:

- Manage the physical loading and unloading of the sensors and goods.
- Constraint: Only touch the status of the trip

3. QA / Compliance Officer:

- Investigate temperature breaches and make legal decisions on whether a shipment is safe to use.
- Constraint: Have no power to change or delete the raw temperature data that triggered it.

4. System Administrator:

- They manage human access to the platform.
- Constraint: Restricted to ensure faking of temperature data does not take place.

SHIPMENTS			
varchar	shipment_id	PK	
varchar	origin		
varchar	destination		
varchar	status		
float	min_temp_limit		e.g., 2.0
float	max_temp_limit		e.g., 8.0

tracks

SENSORS			
varchar	sensor_id	PK	
varchar	shipment_id	FK	
timestamp	calibration_date		

generates (Timescale Hypertable)

TELEMETRY_DATA			
timestampz	time	PK	Part of Composite PK
varchar	sensor_id	PK,FK	Part of Composite PK
float	temperature		
float	latitude		
float	longitude		

triggers breach

USERS			
uuid	user_id	PK	
varchar	name		
enum	role		QA_OFFICER, MANAGER, etc.
varchar	email		
varchar	password_hash		
timestampz	created_at		

acknowledges / resolves

COMPLIANCE_EXCURSIONS			
uuid	excursion_id	PK	
varchar	sensor_id	FK	
timestampz	breach_time		
float	recorded_temp		
uuid	acknowledged_by	FK	Links to USERS
text	resolution_note		

1. Shipment Management Module

Functional Requirements

- Register shipment details
- Register IoT devices
- Configure temperature limits
- Update shipment status

2. IoT Data Simulation Module

Functional Requirements

- Generate simulated temperature readings
- Associate simulated data with a device ID
- Send telemetry data periodically
- Simulate temperature variations for testing excursions

3. Telemetry Ingestion Module

Functional Requirements

- Receive telemetry data via API
- Validate device identity
- Validate shipment association
- Store temperature readings

4. Excursion Detection Module

Functional Requirements

- Compare temperature readings with configured limits
- Detect temperature excursions
- Create excursion records
- Trigger alerts

5. Alert Management Module

Functional Requirements

- Generate alerts when excursions occur
- Store alert logs
- Notify logistics managers
- Display alerts on dashboard

6. Reporting Module

Functional Requirements

- Generate temperature history reports
- Generate excursion reports
- Allow compliance officers to view reports

7. Dashboard Module

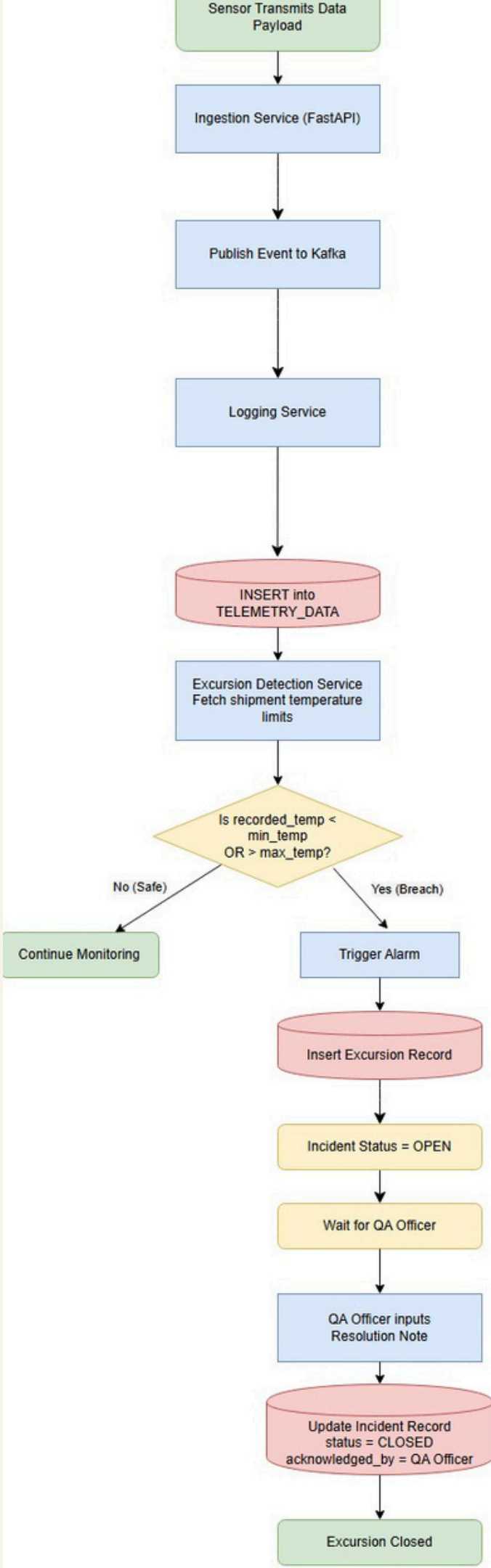
Functional Requirements

- View real-time temperature data
- View alerts
- View excursion events
- Generate reports

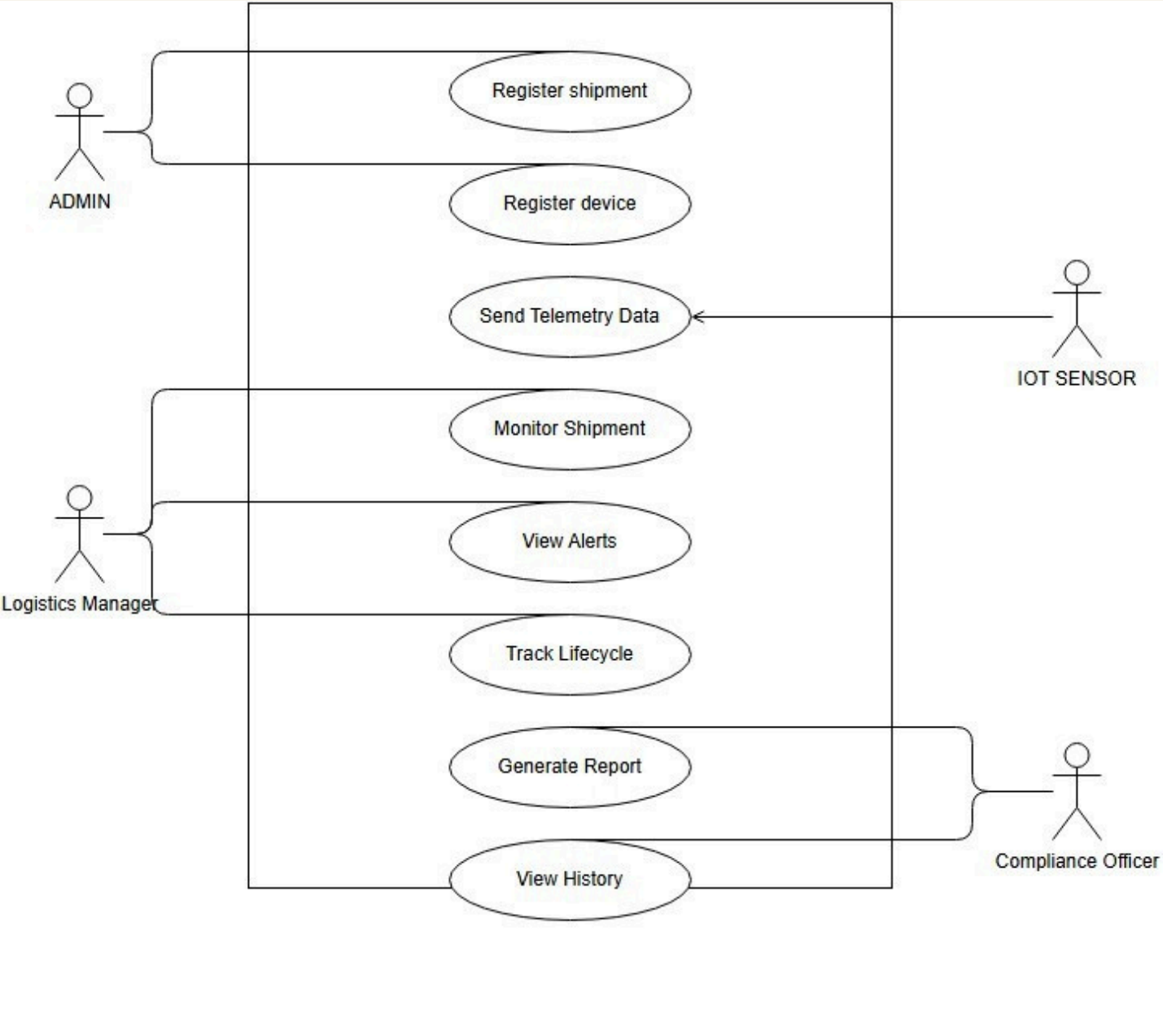
2. Non-Functional Requirements

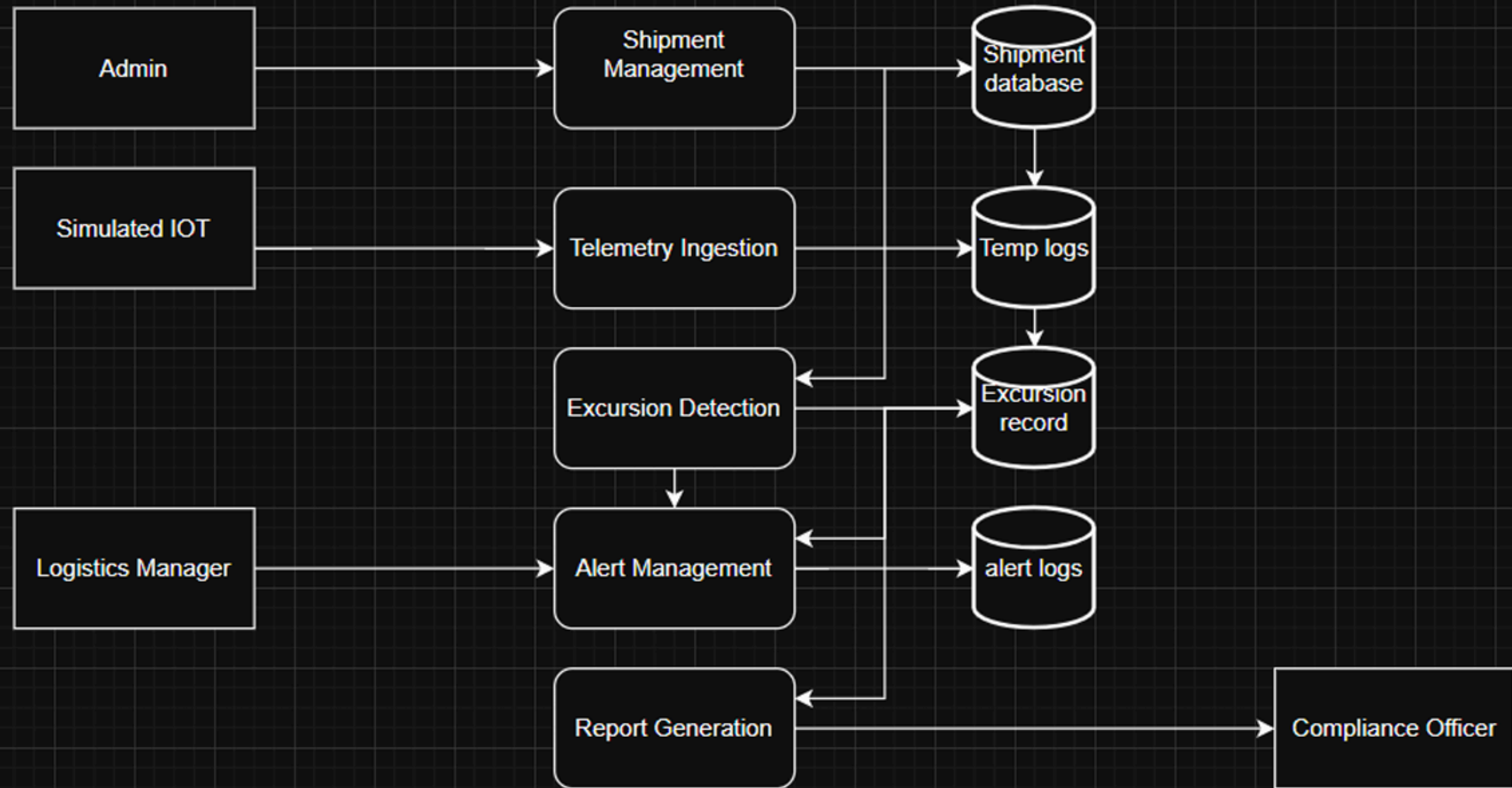
- Scalability
- Reliability
- Security
- Performance
- Maintainability

Activity diagram :



Usecase diagram:





Data Flow Diagram :

